

Activation Study of Precipitated Iron Fischer–Tropsch Catalysts[†]

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Slurry phase Fischer–Tropsch synthesis (FTS) was conducted with two precipitated iron catalysts (100 Fe/3.6 Si/0.71 K and 100 Fe/4.4 Si/1.0 K, atomic percent relative to Fe) at 543 K, 1.31 MPa, and a synthesis gas ($H_2/CO = 0.7$) space velocity of $3.1 \text{ normal L h}^{-1} (\text{g of Fe})^{-1}$. The impact of activation gas (CO , $H_2/CO = 0.7$, or $H_2/CO = 0.1$), temperature (543 or 573 K), and pressure (1.31 or 0.10 MPa) on the long-term ($>500 \text{ h}$) activity and selectivity of the catalysts was explored. Pretreatment with CO under the conditions employed gave highly active and stable catalysts. Catalyst performance when synthesis gas activation was used was found to be dependent upon the partial pressure of hydrogen in the activating gas, with low hydrogen partial pressures resulting in the highest catalyst activity. X-ray diffraction results indicate that carbon monoxide activations and synthesis gas activations with low hydrogen partial pressure result in the formation of the carbides $\chi\text{-Fe}_5\text{C}_2$ and $\epsilon'\text{-Fe}_{2.2}\text{C}$, while activation with synthesis gas with high hydrogen partial pressure results in the formation of only Fe_3O_4 . It was found that treating the 100 Fe/3.6 Si/0.71 K catalyst activated with synthesis gas at 1.31 MPa and 543 K with carbon monoxide caused the activity to increase dramatically and the Fe_3O_4 to be partially converted to iron carbides. It is concluded that Fe_3O_4 is relatively inactive for FTS, while the presence of some bulk iron carbide is necessary for high FTS activity to be achieved.

Introduction

The activation procedure used for iron-based Fischer–Tropsch synthesis (FTS) catalysts has a great influence on their activity and selectivity.¹ Reduction of fused magnetite catalysts with hydrogen at high temperatures ($>673 \text{ K}$) and high linear flow rates is necessary to produce a substantial surface area necessary for good activity; however, reduction with synthesis gas or carbon monoxide is ineffective.² In contrast, activation of precipitated iron catalysts is not understood as well. Precipitated catalysts have been reported to be successfully activated with carbon monoxide,^{3,4} hydrogen,^{2,5} or synthesis gas.^{6–8}

Several research groups have reported that pretreatment of precipitated iron oxide with carbon monoxide is the preferred activation procedure. Bukur et al. have

reported that pretreatment of doubly promoted Fe/Cu/K catalysts with carbon monoxide or synthesis gas ($H_2/CO \sim 0.7$ or 2.0) results in higher initial activity and a heavier product than activation with hydrogen; however, hydrogen pretreatments were reported to result in more stable catalysts.^{4,5} Pennline et al. found that activation of a precipitated iron–manganese catalyst with carbon monoxide was superior to activation with hydrogen or synthesis gas at 15 MPa and 548 K; furthermore, they report little temperature or pressure effects when activating with carbon monoxide.⁹ During the initial German work on FTS, the Kaiser Wilhelm Institute favored pretreating catalysts with carbon monoxide at subatmospheric pressure.¹⁰ In addition, Huang et al. have reported that an ultrafine iron oxide catalyst pretreated with carbon monoxide has activity superior to that pretreated with hydrogen or synthesis gas ($H_2/CO = 1.0$) at 533 K and 0.79 MPa.¹¹

Although pretreatment of precipitated iron catalysts with carbon monoxide leads to active catalysts, hydrogen pretreatment is more favorable on an industrial scale, because it is less expensive and easier to purify. In general, pure hydrogen can be prepared from synthesis gas using a membrane separation process that allows only hydrogen to pass; however, this separation will not lead to pure carbon monoxide, but instead will produce a carbon monoxide rich synthesis gas. SASOL utilizes hydrogen pretreatment of precipitated iron catalysts used in its commercial FT reactors.² In

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(1) Anderson, R. B. *The Fischer–Tropsch Synthesis*; Academic Press: Orlando, FL, 1984.

(2) Dry, M. E. In *Catalysis—Science and Technology*; Anderson, J. R., Boudart, M., Eds.; Springer-Verlag: New York, 1981; Vol. 1, pp 159–255.

(3) O'Brien, R. J.; Xu, L.; Milburn, D. R.; Li, Y.-X.; Klabunde, K. J.; Davis, B. H. *Top. Catal.* **1995**, 2, 1.

(4) Bukur, D. B.; Lang, X.; Rossin, J. A.; Zimmerman, W. H.; Rosynek, M. P.; Yeh, E. B.; Li, C. *Ind. Eng. Chem. Res.* **1989**, 28, 1130.

(5) Bukur, D. B.; Koranne, M.; Lang, X.; Rao, K. R. P. M.; Huffman, G. P. *Appl. Catal.* **1995**, 126, 85.

(6) Kölbl, H.; Ralek, M. *Catal. Rev.-Sci. Eng.* **1980**, 21, 225.

(7) Kuo, J. C. W. *Slurry Fischer–Tropsch/Mobil Two-Stage Process of Converting Syngas to High Octane Gasoline*; DOE/PC/30022-10, Final Report, June 1983.

(8) Kuo, J. C. W. *Two Stage Process for Conversion of Synthesis Gas to High Quality Transportation Fuels*; DOE/PC/60019-9, Final Report, Oct 1985.

(9) Pennline, H. W.; Zaroachak, M. F.; Stencel, J. M.; Diehl, J. R. *Ind. Eng. Chem. Res.* **1987**, 26, 595.

(10) Report on the Petroleum and Synthetic Oil Industry of Germany, Ministry of Fuel and Power; His Majesty's Stationary Office, London, 1946; pp 96–100.

(11) Huang, C.-S.; Xu, L.; Davis, B. H. *Fuel Sci. Technol. Int.* **1993**, 11, 639.

addition, much of the early German work utilized hydrogen pretreatment, i.e., Ruhrchemie AG, IG Farbenindustrie AG, and Lurgi.¹⁰ Hydrogen pretreatment has the disadvantage that the metallic iron that is formed is susceptible to sintering. High flow rates are needed to keep the partial pressure of water low, and low temperatures are necessary to ensure that a high surface area is maintained.^{2,5}

The development of a pretreatment procedure utilizing synthesis gas that would consistently lead to active catalysts would be attractive, because pretreatments could be performed with the same gas that is used in the FTS, thereby eliminating the need for a pure hydrogen or carbon monoxide stream. Kölbel and Ralek⁶ and Mobil^{7,8} have reported high activity and good stability with precipitated Fe/Cu/K catalysts in laboratory and demonstration plant slurry reactors using *in situ* synthesis gas activation. Pretreatment conditions appear to be crucial in determining the ultimate FT activity of synthesis gas activated catalysts. Kölbel and Ralek report that the temperature of the synthesis gas pretreatment is critical to the final activity and selectivity of the catalyst; too high a temperature will result in "supercarbonization" and too low a temperature will prevent activation.⁶ In addition, pressure may also affect the ultimate activity; many of the German companies preferred activation at a low pressure when using synthesis gas pretreatments.¹⁰

The catalyst phase that exists is strongly dependent on the activation procedure employed. Activation with hydrogen, carbon monoxide, or synthesis gas generally results in the rapid formation of Fe₃O₄.³ With additional time, the Fe₃O₄ is converted to metallic iron in the case of hydrogen pretreatment or various iron carbides (usually χ -Fe₅C₂ or ϵ' -Fe_{2.2}C) with carbon monoxide or synthesis gas pretreatment.^{1-3,5} During synthesis, metallic iron is rapidly converted to iron carbide and with longer times on stream the iron carbide may be oxidized to Fe₃O₄.^{1-3,5} Several studies have been made to correlate catalyst composition with activity.^{3,12-15} Some reports claim iron carbides are active for FTS¹² while others claim an oxide species is responsible for activity.¹⁶ In contrast, it has also been reported that a correlation between catalyst activity and bulk phase composition cannot be made.¹⁻³

The present study is aimed at providing a better understanding of the relationship between pretreatment, FTS activity, and the phase composition of precipitated iron catalysts used in a slurry reactor. A comparison is made between pretreatment with carbon monoxide and synthesis gas (H₂/CO = 0.7 or 0.1) with particular attention being paid to the effect of temperature and pressure used during the activation procedure on the resulting FTS activity.

Experimental Section

Catalysts were prepared by continuous precipitation of amorphous ferric oxyhydroxide from aqueous iron(III) nitrate nonahydrate (1.17 M) and concentrated ammonium hydroxide

(15.6 M) in a continuous stirred tank reactor at pH 9.5. Hydrolyzed tetraethyl orthosilicate was added to the iron(III) nitrate solution to give the desired level of silicon. The catalyst slurry was filtered continuously with one or more 6 in. rotary drum vacuum filters. The resulting catalyst cake was washed and filtered twice with a volume of distilled deionized water equal to the volume of the previous filtrate. A 5 kg catalyst batch was precipitated in this manner, oven-dried in flowing air at 393 K, and crushed to -300 mesh. The composition of this catalyst in terms of atomic percent relative to Fe is 100 Fe/3.6 Si. During loading into the slurry reactor, the catalyst was treated with sufficient potassium *tert*-butoxide to make the composition 100 Fe/3.6 Si/0.71 K. A large 75 kg catalyst batch was also prepared; however, this catalyst was prepared by adding potassium nitrate to the slurry, spray-dried, and calcined in air at 623 K. The composition of this catalyst was 100 Fe/4.4 Si/1.0 K.

All FTS runs were conducted in a 1 L autoclave operating as a continuous stirred tank reactor (CSTR). Catalysts were suspended in Ethylflo 164 hydrocarbon oil (Ethyl Corp.), which is reported to be a C₃₀ 1-decene homopolymer. All catalyst loadings were 10 wt % unless otherwise noted. A schematic of the reactor system is shown in Figure 1. Hydrogen and carbon monoxide feed gas flow rates were controlled by two mass flow controllers (Brooks Instruments) with the resulting synthesis gas composition regulated by adjusting the flow rate of the appropriate gas. The synthesis gas, after passing through a 2 L mixing vessel, was delivered to the catalyst slurry through a dip tube that extended to below the impeller blade. The reactor effluent exited the reactor and passed sequentially through two traps maintained at 333 and 273 K. Accumulated reactor wax was removed daily through a tube fitted with a porous metal filter (0.5 μ m). Uncondensed effluent was sent to an on-line Carle gas analyzer for determination of carbon monoxide, hydrogen, carbon dioxide, methane, and C₂-C₄ alkanes and alkenes. An on-line Hewlett-Packard 5790 GC equipped with a Porpack-Q column was utilized for C₄-C₉ quantification. Liquid samples were analyzed with a Hewlett-Packard 5890 GC equipped with a 60 m DB-5 capillary column. The reactor was also equipped with a tube that extended below the liquid level, which permitted catalyst slurry samples to be withdrawn periodically.

Catalysts were pretreated with carbon monoxide or activated with synthesis gas with a hydrogen/carbon monoxide ratio of 0.7 or 0.1. In general, the activation gas flow was started at ambient conditions, the necessary reactor pressure was set, and the reactor was ramped to the desired temperature at 2 K min⁻¹. After the activation temperature was reached, the conditions were maintained for 24 h. Following the activation treatment, the reactor was brought to reaction conditions: hydrogen/carbon monoxide ratio of 0.7, 3.1 normal L h⁻¹ (g of Fe)⁻¹ (normal liters corresponds to 273 K, 0.10 MPa), 543 K, and 1.31 MPa.

Catalyst samples were Soxhlet extracted according to the method of McCartney et al.¹⁷ to remove accumulated wax and then analyzed by X-ray diffraction. Powder X-ray diffraction patterns of the catalysts were obtained using a Philips APD X-ray diffraction spectrometer equipped with a Cu anode and Ni filter operated at 40 kV and 20 Ma (Cu K α = 1.5418 Å). Iron phases were identified by comparing diffraction patterns of the catalyst samples with those in the standard powder X-ray diffraction file compiled by the Joint Committee on Powder Diffraction Standards published by the International Center for Diffraction Data.

Results and Discussion

Catalyst Activation. Early in our efforts to develop an effective activation procedure for precipitated iron

(12) Amelse, J. A.; Butt, J. B.; Schwartz, L. H. *J. Phys. Chem.* **1978**, *82*, 558.

(13) Niemantsverdriet, J. W.; van der Kraan, A. M.; van Dijk, W. L.; van der Baan, H. S. *J. Phys. Chem.* **1980**, *84*, 3363.

(14) Butt, J. B. *Catal. Lett.* **1990**, *7*, 61.

(15) Butt, J. B. *Catal. Lett.* **1990**, *7*, 83.

(16) Reymond, J. P.; Mériaudeau, P.; Teichner, S. J. *J. Catal.* **1982**, *75*, 39.

(17) McCartney, J. T.; Hofer, L. J. E.; Seligman, B.; Lecky, J. A.; Peebles, W. C.; Anderson, R. B. *J. Phys. Chem.* **1953**, *57*, 730.

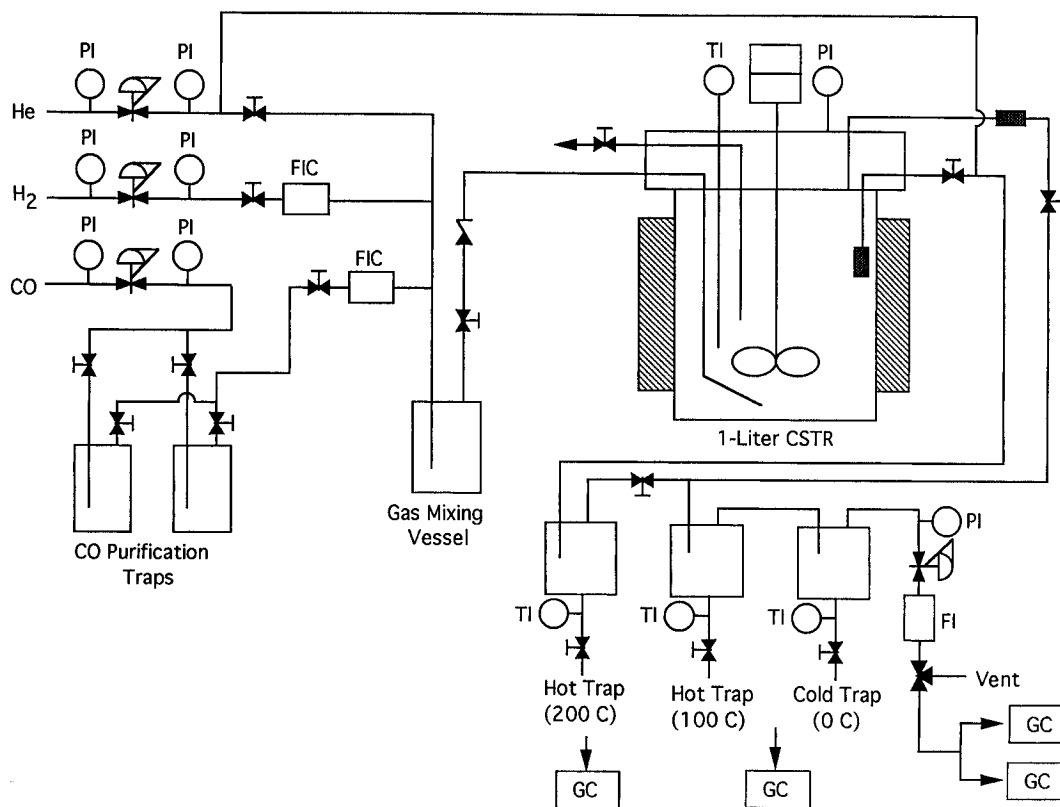


Figure 1. Schematic of the 1 L continuous stirred tank reactor system.

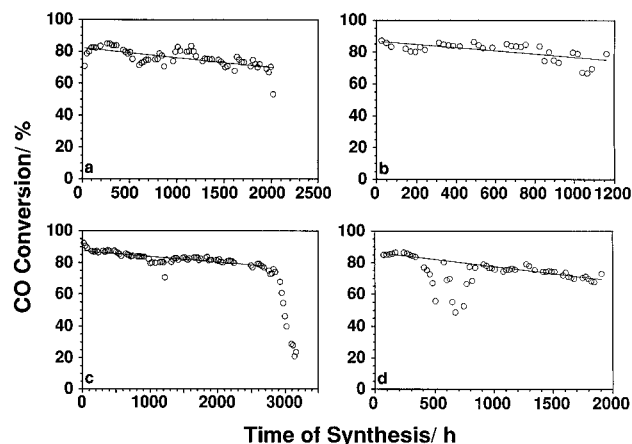


Figure 2. Carbon monoxide conversion for 100 Fe/4.4 Si/1.0 K catalyst pretreated for 24 h with (a) carbon monoxide at 543 K and 1.31 MPa, (b) carbon monoxide at 543 K and 0.10 MPa, (c) synthesis gas ($H_2/CO = 0.7$) at 543 K and 0.10 MPa, and (d) synthesis gas ($H_2/CO = 0.7$) at 573 K and 0.10 MPa.

catalysts, we found that a 24 h treatment with carbon monoxide at 543 K and 1.31 MPa consistently resulted in highly active and stable catalysts (Figures 2a and 3). Initial carbon monoxide conversion for the 100 Fe/3.6 Si/0.71 K and 100 Fe/4.4 Si/1.0 K catalysts activated in this manner were 92% and 85%, respectively, with corresponding deactivation rates of 4.8% and 1.0% per week, respectively. Hydrocarbon productivities typically were in the range of 160–170 g m^{-3} (CO + H_2 feed) with greater than 80% being C_3 and higher. The reproducibility of this activation procedure has facilitated the screening of large numbers of catalysts so that reliable data concerning optimum promoter levels can be determined. Much of this work is directed at developing a process for catalyst preparation that is suitable for scale-up to pilot plant scale.

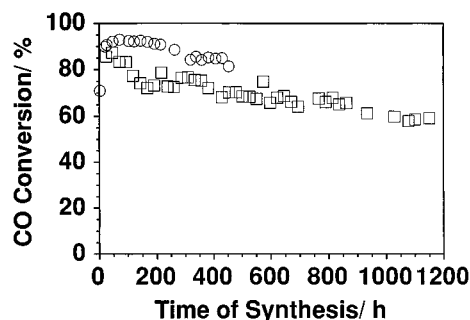


Figure 3. Carbon monoxide conversion for 100 Fe/3.6 Si/0.71 K catalyst pretreated for 24 h with (O) carbon monoxide at 543 K and 1.31 MPa and (□) synthesis gas ($H_2/CO = 0.7$) at 543 K and 0.10 MPa.

The use of pure carbon monoxide as an activation gas is acceptable for small laboratory reactors; however, it is cost prohibitive for a pilot plant. As a result, it is economically attractive and operationally more facile to be able to activate catalysts *in situ* with the same synthesis gas as used during the FTS. Activity and selectivity results of our initial efforts using synthesis gas activation are shown in Table 1. Treatment of either 100 Fe/3.6 Si/0.71 K or 100 Fe/4.4 Si/1.0 K catalysts with synthesis gas ($H_2/CO = 0.7$) at 3.1 normal L h^{-1} (g of Fe) $^{-1}$, 543 K, and 1.31 MPa resulted in poor FTS activity. Carbon monoxide conversions were typically below 30% and did not increase appreciably with time on stream. Kölbel and Ralek have reported successful synthesis gas activations with a precipitated Fe/Cu/K catalyst at 0.8–1.5 MPa and at a “formation” temperature 15–30 K above the FTS temperature (~573 K); however, they report that at too high a temperature carbon will form and at too low a temperature the formation will not take place and the catalyst will be inactive.⁶ In contrast, we have found that

Table 1. Results of Syngas Activations at 1.31 MPa

catalyst: activation conditions:	100 Fe/3.6 Si/0.71 K						100 Fe/4.4 Si/1.0 K	
	H ₂ /CO = 0.7, 1.31 MPa, 543 K		H ₂ /CO = 0.7, 1.31 MPa, 573 K		H ₂ /CO = 0.1, 1.31 MPa, 543 K		H ₂ /CO = 0.7, 1.31 MPa, 543 K	
time on stream, h	46	92	46	119	47	139	48	192
conversion, %								
CO	12	12	15	21	62	60	22	33
H ₂	22	26	21	26	58	54	28	35
CO + H ₂	16	18	18	23	61	57	25	34
H ₂ /CO usage ratio	1.3	1.5	0.98	0.87	0.65	0.63	0.89	0.74
CO ₂ selectivity, mol %	37	43	35	34	46	46	38	43
hydrocarbon productivity								
g of HC (normal m of feed gas) ⁻³	27	25	37	51	126	120	52	70
g of C ₃ ⁺ (normal m of feed gas) ⁻³	22	20	31	44	108	103	44	59
g of HC (g of Fe) ⁻¹ h ⁻¹	0.084	0.078	0.11	0.16	0.39	0.37	0.16	0.21
alkene selectivity, mol %								
C ₂	58	74	62	67	23	16	75	63
C ₃	92	91	84	89	69	62	83	81
C ₄	83	84	81	84	76	76	86	84

Table 2. FTS Results for Catalysts at High Activity

	pretreatment							
	CO/543 K/1.31 MPa		CO/543 K/0.10 MPa		H ₂ /CO = 0.7/ 543 K/0.10 MPa		H ₂ /CO = 0.7/ 573 K/0.10 MPa	
time on stream, h	336	1992	360	1016	312	2852	312	1836
space velocity, normal L h ⁻¹ (g of Fe) ⁻¹	3.4	3.4	3.1	3.1	3.1	3.1	3.1	3.1
conversion, %								
CO	84	74	84	79	87	74	84	68
H ₂	76	65	75	73	77	67	76	67
CO + H ₂	81	70	80	76	83	71	81	67
WGS								
CO ₂ , mol %	46	42	45	44	50	46	47	44
H ₂ /CO usage	0.63	0.61	0.63	0.65	0.62	0.64	0.62	0.68
K _{app}	14	7.2	16	12	40	30	16	4.4
productivity								
g of HC (normal m of feed gas) ⁻³	167	159	168	162	162	146	165	140
g of C ₃ ⁺ (normal m of feed gas) ⁻³	142	132	142	134	130	110	136	116
g of HC (g of Fe) ⁻¹ h ⁻¹	0.62	0.58	0.52	0.50	0.50	0.45	0.50	0.42
selectivity, wt % (alkene)								
C ₁	8.1	8.7	8.2	8.7	11	15	9.6	8.3
C ₂ -C ₄	26 (57)	28 (61)	27 (57)	29 (57)	32 (48)	37 (53)	30 (54)	33 (65)
C ₅ -C ₁₁	34 (58)	27 (66)	36 (60)	32 (60)	46 (54)	34 (56)	34 (59)	40 (68)
C ₁₂ ⁺	32	36	29	31	11	15	26	19

increasing the activation temperature to 573 K while maintaining the pressure at 1.31 MPa has little effect on conversion during synthesis with catalysts activated with synthesis gas.

During the Reichsamt comparative experiments conducted in Germany in 1943, the Kaiser Wilhelm Institute, Brabag, and Rheinpreussen all activated their iron catalysts with synthesis gas or water gas at atmospheric or subatmospheric pressure.¹⁰ To explore the possible effect of pressure on synthesis gas activation, both 100 Fe/3.6 Si/0.71 K (Figure 3) and 100 Fe/4.4 Si/1.0 K (Figure 2c) catalysts were treated with synthesis gas (H₂/CO = 0.7) at 3.1 normal L h⁻¹ (g of Fe)⁻¹, 0.10 MPa, and 543 K for 24 h. A dramatic increase in activity was observed as compared to when the same activation was conducted at 1.31 MPa. For both the 100 Fe/3.6 Si/0.71 K and 100 Fe/4.4 Si/1.0 K catalysts, the initial carbon monoxide conversion was greater than 85%, which is comparable to that obtained with carbon monoxide pretreatment at 543 K and 1.31 MPa. In fact, the 100 Fe/4.4 Si/1.0 K catalyst had an initial conversion of 92% and displayed remarkable stability, deactivating at an average rate of only 0.59% per week for over 2700 h. In addition, the 100 Fe/4.4 Si/1.0 K catalyst was also activated with synthesis gas (H₂/CO = 0.7) at 573 K and 0.10 MPa. As was found when activation was done with synthesis gas at 1.31 MPa, increasing the temperature

had little effect on the initial carbon monoxide conversion (Figure 2d); however, the deactivation rate was higher at 1.6% per week.

It is interesting to note that no such pressure dependence was observed with carbon monoxide pretreatment (Figure 2b). Pretreating the 100 Fe/4.4 Si/1.0 K catalyst with carbon monoxide for 24 h at 543 K and 1.31 MPa resulted in an initial conversion slightly lower than when activation was done at 0.10 MPa; however, after 72 h at synthesis conditions, the conversions were essentially the same in both cases.

Selectivity data at high conversion for the 100 Fe/4.4 Si/1.0 K catalyst activated with carbon monoxide or synthesis gas are presented in Table 2. In general, carbon monoxide activations resulted in the production of less methane and more liquid products than activations with synthesis gas. At comparable conversions, the catalyst activated with carbon monoxide at either 543 K/1.31 MPa or 543 K/0.10 MPa produced up to twice as much C₁₂⁺ as when activated with synthesis gas at 0.10 MPa. In contrast, Bukur et al. reported similar selectivities for a doubly promoted Fe/Cu/K catalyst that was activated with carbon monoxide or with synthesis gas.⁵

The fact that pretreatment with carbon monoxide at either 0.10 or 1.31 MPa results in similar FTS activity suggests that the pressure dependence of synthesis gas

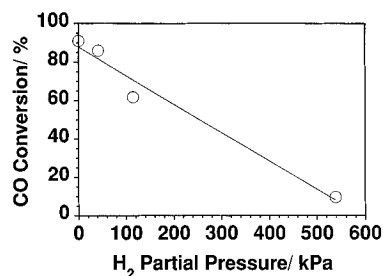


Figure 4. Initial carbon monoxide conversion as a function of the partial pressure of hydrogen in the pretreatment gas for the 100 Fe/3.6 Si/0.71 K catalyst.

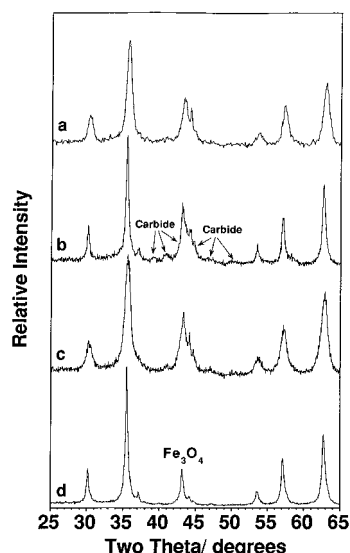


Figure 5. X-ray diffraction results for 100 Fe/3.6 Si/0.71 K catalyst following 24 h pretreatment with (a) carbon monoxide at 543 K and 1.31 MPa, (b) synthesis gas ($H_2/CO = 0.7$) at 543 K and 0.10 MPa, (c) synthesis gas ($H_2/CO = 0.1$) at 543 K and 1.31 MPa, and (d) synthesis gas ($H_2/CO = 0.7$) at 573 K and 1.31 MPa.

activation is due to the presence of hydrogen. To test this possibility, the 100 Fe/3.6 Si/0.71 K catalyst was treated with a synthesis gas low in hydrogen ($H_2/CO = 0.1$) for 24 h at 2.1 normal L h^{-1} (g of Fe) $^{-1}$, 543 K, and 1.31 MPa. The initial carbon monoxide conversion was approximately 65%, which is intermediate between activation with pure carbon monoxide and activation with synthesis gas with a hydrogen/carbon monoxide ratio of 0.7. As shown in Figure 4, there is a clear correlation between the partial pressure of hydrogen in the activating gas and the carbon monoxide conversion obtained during the early synthesis period.

Catalyst Characterization. Catalyst slurry samples were removed from the reactor for the series of runs with the 100 Fe/3.6 Si/0.71 K catalyst. X-ray diffraction analysis of the catalyst following pretreatment with CO (Figure 5a) or synthesis gas at 0.10 MPa (Figure 5b) shows the presence of Fe_3O_4 and the carbides χ - Fe_5C_2 and ϵ' - $Fe_{2.2}C$. In addition, X-ray diffraction data for the catalyst activated with hydrogen-lean synthesis gas ($H_2/CO = 0.1$) (Figure 5c) showed Fe_3O_4 and small amounts of iron carbides. In contrast, X-ray diffraction analysis of the catalyst activated with synthesis gas at 1.31 MPa showed only Fe_3O_4 (Figure 5d).

As discussed above, the carbon monoxide conversion of the 100 Fe/3.6 Si/0.71 K catalyst activated with synthesis gas ($H_2/CO = 0.7$) at 1.31 MPa and 543 K

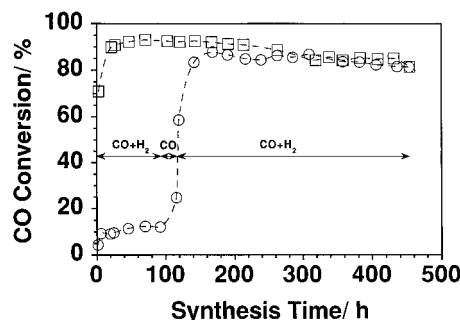


Figure 6. Carbon monoxide conversion for 100 Fe/3.6 Si/0.71 K catalyst (○) activated with synthesis gas at 543 K and 1.31 MPa, followed by a carbon monoxide treatment at 543 K and 1.31 MPa after 96 h of synthesis and (□) pretreated with carbon monoxide for 24 h at 543 K and 1.31 MPa.

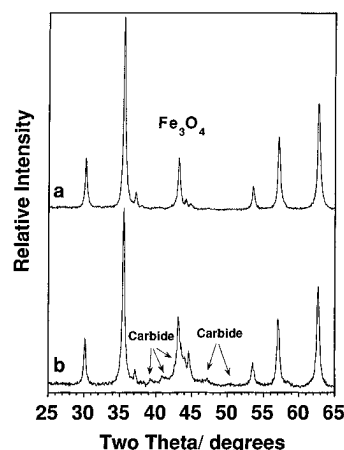
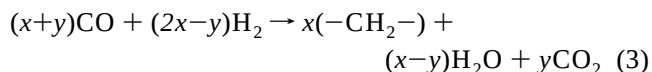
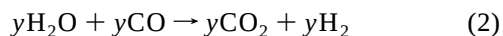


Figure 7. X-ray diffraction results for the 100 Fe/3.6 Si/0.71 K catalyst (a) after 96 h of exposure to synthesis gas at 543 K and 1.31 MPa and (b) followed by a 22 h exposure to carbon monoxide at 543 K and 1.31 MPa

never rose above 12%. After 96 h of synthesis, the hydrogen flow was stopped and the catalyst was treated with carbon monoxide under the same operating conditions for 22 h. Following the carbon monoxide treatment, the hydrogen flow was resumed and the activity of the catalyst was found to rapidly increase (Figure 6). The ultimate activity was similar to that obtained for the carbon monoxide pretreated catalyst. X-ray diffraction analysis showed that the catalyst consisted of only Fe_3O_4 prior to the carbon monoxide treatment; however, following the carbon monoxide treatment, the catalyst consisted of a mixture of Fe_3O_4 , χ - Fe_5C_2 , and ϵ' - $Fe_{2.2}C$ (Figure 7). These results are consistent with the Fe_3O_4 phase being relatively inactive for FTS, while χ - Fe_5C_2 and ϵ' - $Fe_{2.2}C$ are necessary for high FTS activity. This does not necessarily imply that χ - Fe_5C_2 and ϵ' - $Fe_{2.2}C$ are active phases for FTS; in fact, previous works have shown that there is no correlation between the amount of bulk iron carbide present and the activity of the catalyst.^{2,3} It is possible, however, that the presence of a bulk iron carbide is necessary to serve as a support for an active surface species. In considering the above results and discussion, one should keep in mind that the characterization is based upon XRD; therefore, only bulk phases are detected.

The overall FTS with iron catalysts can be approximated as the sum of the hydrogenation of carbon monoxide to methylene groups and the water-gas shift reaction.



For iron-based catalysts, whether the catalyst exists as a carbide or as an oxide depends on the reducing behavior of the atmosphere present in the reactor. During FTS, the catalyst is exposed to an atmosphere of hydrogen, carbon monoxide, and the reaction products (water and carbon dioxide). It is well-known that water and carbon dioxide will oxidize metallic iron and iron carbides to Fe_3O_4 .^{1,2} Thermodynamic data on the metallic iron/iron oxide system show that at a $\text{H}_2\text{O}/\text{H}_2$ ratio >0.03 or a CO_2/CO ratio >2.8 oxidizing conditions exist and Fe_3O_4 is the stable phase compared to metallic iron.¹⁸ Equation 3 can be used to calculate the $\text{H}_2\text{O}/\text{H}_2$ and CO_2/CO ratios based on the carbon monoxide and hydrogen conversions. During activation of the 100 Fe/3.6 Si/0.71 K catalyst with synthesis gas at 1.31 MPa, the $\text{H}_2\text{O}/\text{H}_2$ ratio was 0.16 and the CO_2/CO ratio was 0.022. The water level clearly was high enough to make Fe_3O_4 the thermodynamically stable phase during the activation, which would explain why no iron carbides could be detected by XRD. This conclusion requires that the formation of iron carbides results from an iron species that has a lower valency than is present in Fe_3O_4 . In contrast, synthesis gas activation of the same catalyst at 0.10 MPa had a much lower $\text{H}_2\text{O}/\text{H}_2$ ratio of 0.0070 and a CO_2/CO ratio of 0.90. These water and carbon dioxide levels were low enough to enable the

catalyst to be reduced to form active iron carbides. It appears that by decreasing the hydrogen partial pressure, reaction 1 is suppressed enough during the activation that active iron carbides can be formed. An alternate method of activation with synthesis gas would be to operate at high linear flow rates (low conversion) to suppress water formation and to rapidly remove it from the reactor, as is usually done when activation is done with pure hydrogen.^{2,5}

Conclusions

CO pretreatment of precipitated Fe/Si/K catalysts consistently results in high FTS activity because active carbide phases are readily formed. In the case of syngas activation, there is a relationship between the hydrogen partial pressure in the activation gas and the initial FTS activity. During activation with syngas, the catalyst is exposed to carbon dioxide and water as well as carbon monoxide and hydrogen. It has been found that during activations with high partial pressures of hydrogen, enough water is formed to prevent the catalyst from being reduced to active iron carbide phases and only relatively inactive Fe_3O_4 is formed. The results of this study indicate that the formation of an iron carbide ($\chi\text{-Fe}_5\text{C}_2$ and/or $\epsilon'\text{-Fe}_{2.2}\text{C}$) is necessary for high FTS activity; however, on the basis of previous Mössbauer spectroscopy experiments, the activity of iron catalysts is not related to the amount of bulk iron carbide present.³ These results indicate that a layer of surface carbide may be responsible for FTS activity.

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(18) Reed, T. B. *Free Energy of Formation of Binary Compounds: An Atlas of Charts for High-Temperature Chemical Calculations*; MIT Press: Cambridge, MA, 1971.